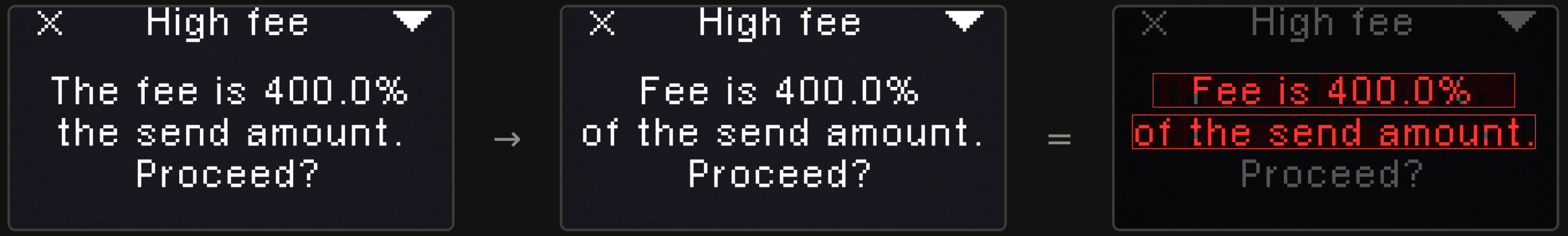
BITBOX-E2E

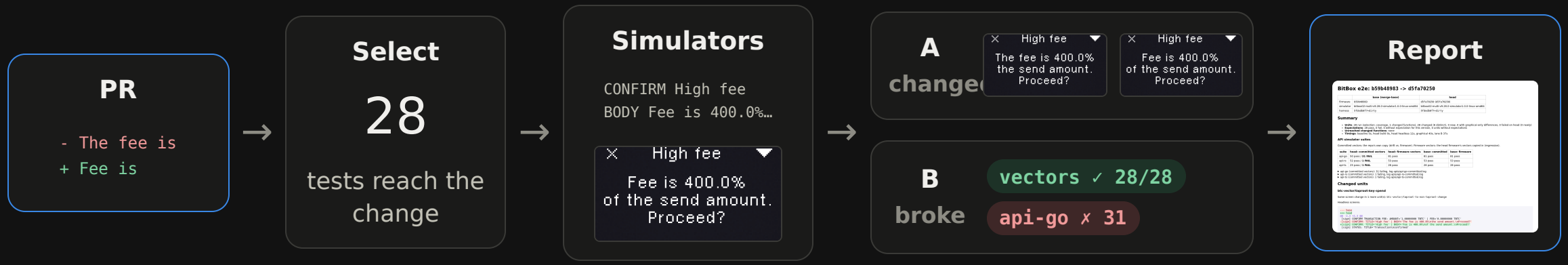
Open a PR.
See what changed on the device.



master · your PR · changed

`automatic` `firmware + API + app + Rabby + WalletConnect` `real simulator pixels` `~90 s per PR`

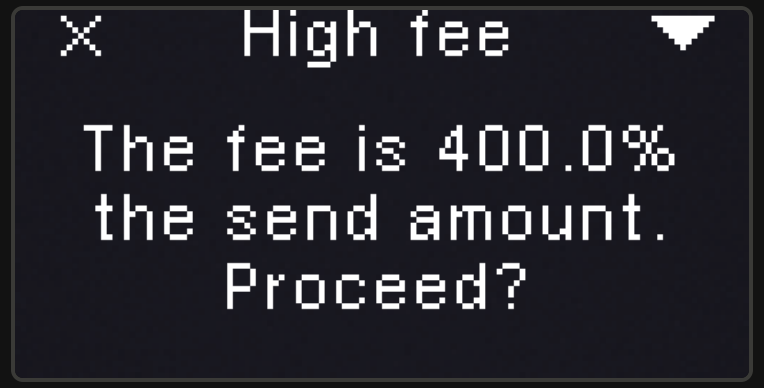
HOW IT RUNS



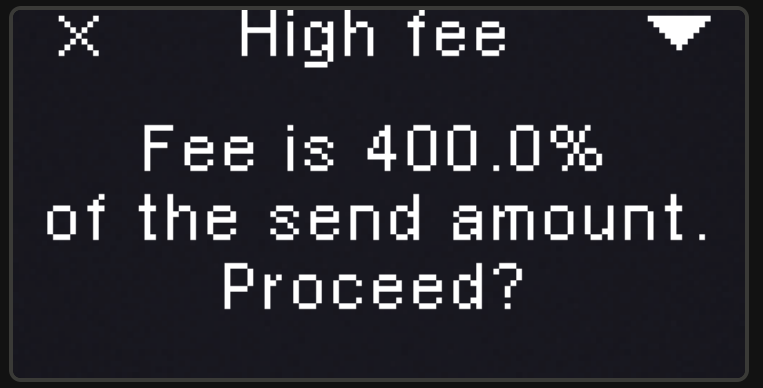
Baseline of master is stored once, PRs run only their own side

Expected screens optional; approved screens become checks

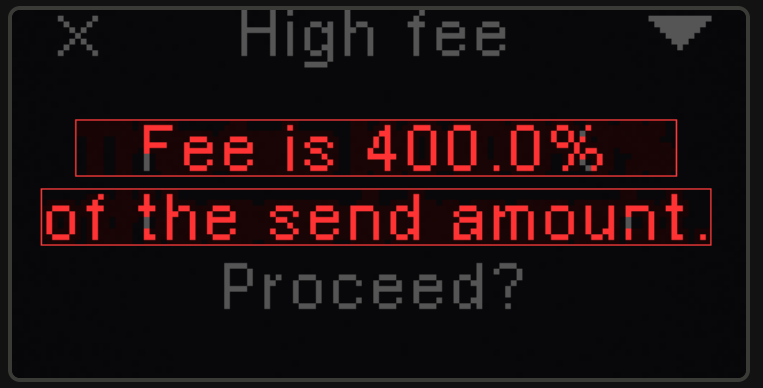
DEVICE · CHANGED SCREEN · FIRMWARE D5FA70250



base b59b48983



head d5fa70250



changed lines in red

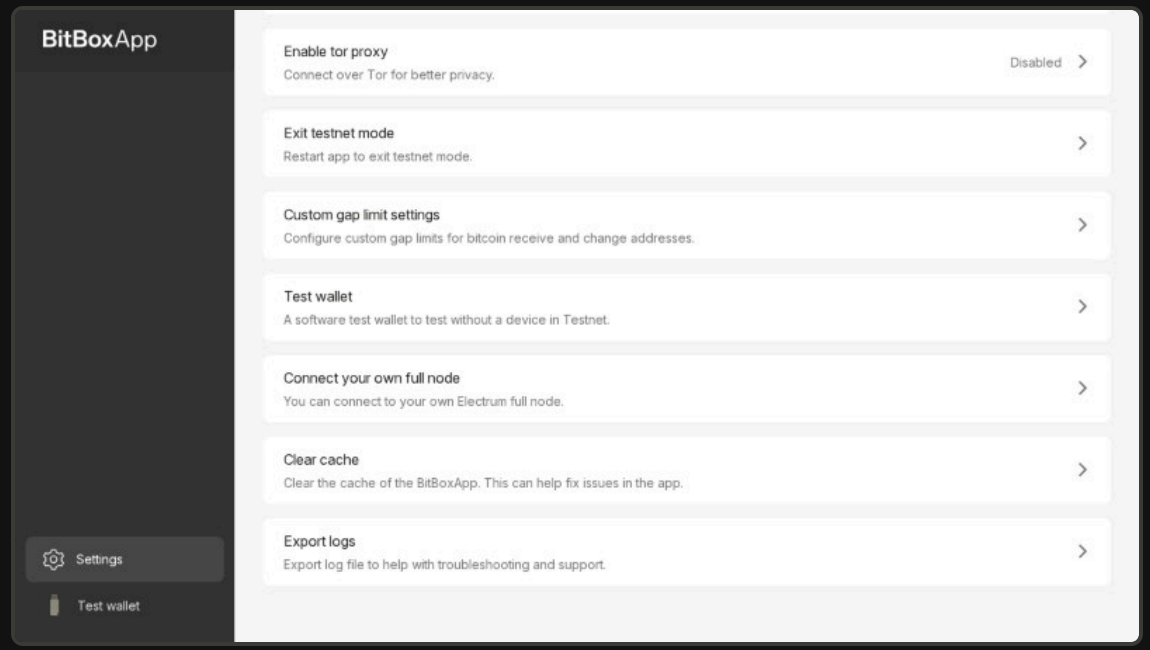
```
btc-vector/taproot-key-spend · CONFIRM "High fee"
- The fee is 400.0%\nthe send amount.\nProceed?
+ Fee is 400.0%\nof the send amount.\nProceed?
```

DEVICE · NEW SCREEN IN A FLOW · ERC-20 TO UNKNOWN CONTRACT

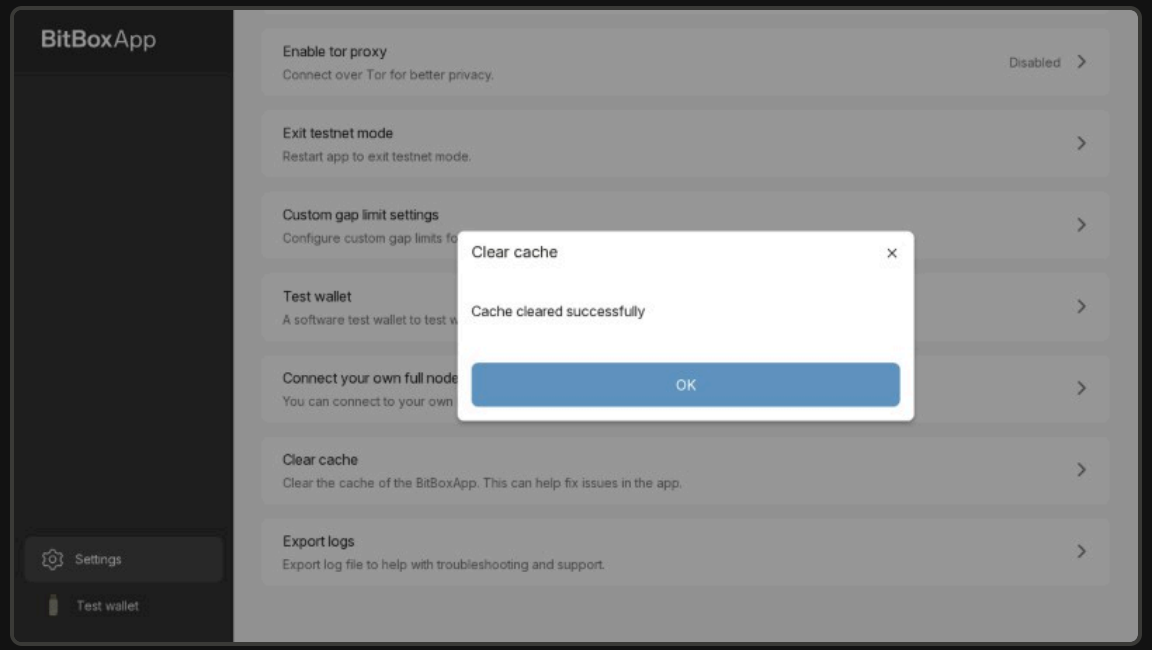


`+ Token contract 0x9c23…47C1` frames aligned by content; setup screens (pairing, seed) left out

BITBOXAPP · WALLET-APP F1104A39D · TESTS/CLEAR-CACHE.TEST.TS



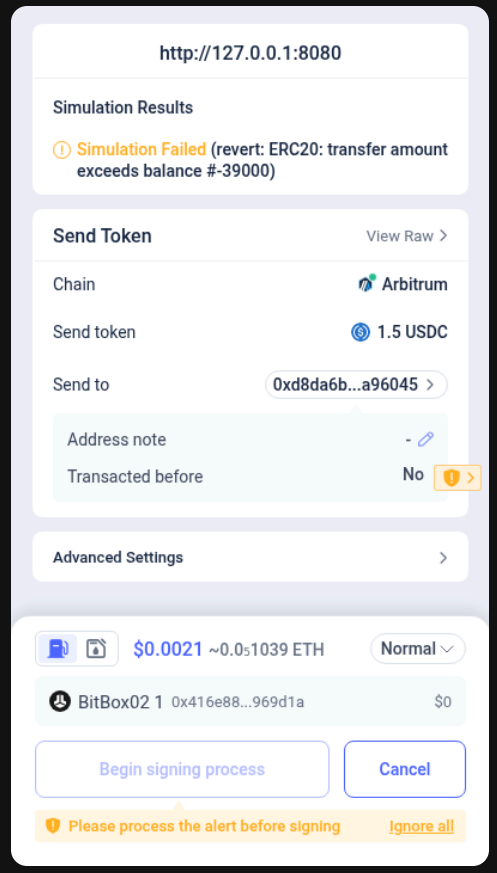
base f1104a39d^ · after "Clear Cache": dialog just closes



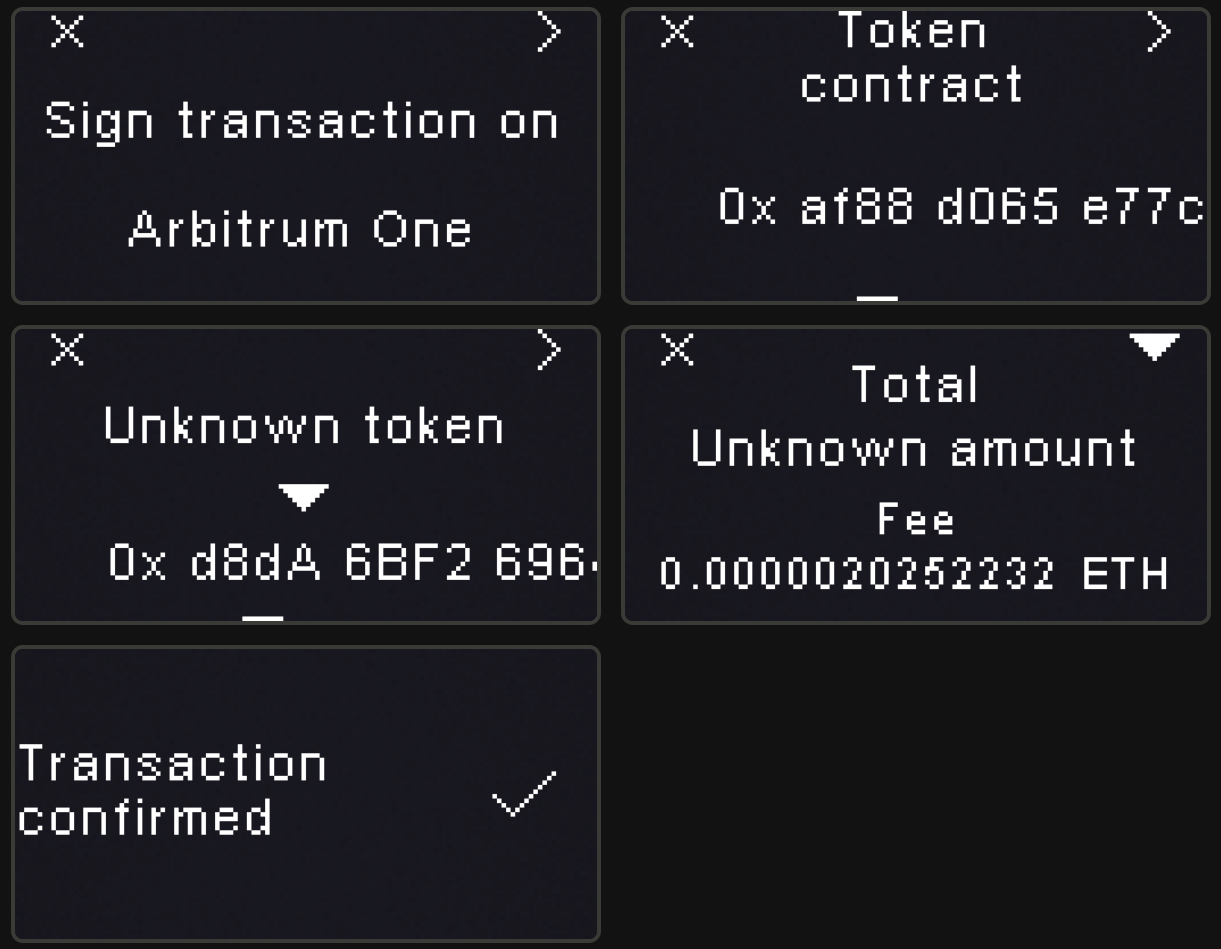
head f1104a39d · new confirmation

`same test step, both refs (from Playwright trace)` `real simulator as the device` `1/1 → 1/1 pass`

RABBY V0.94.15, UNMODIFIED · ERC-20 TRANSFER ON ARBITRUM



Rabby approval



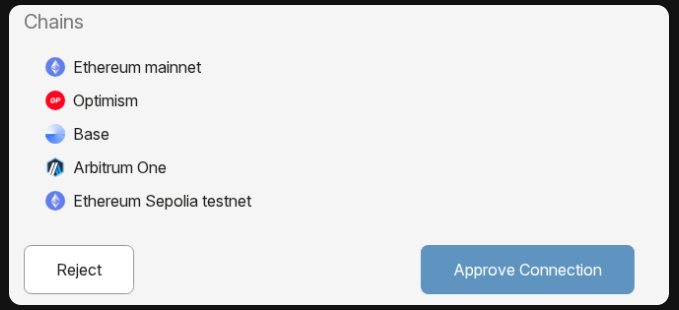
device, same step, in order

test dapp → Rabby → fake BitBoxBridge :8178 → simulator

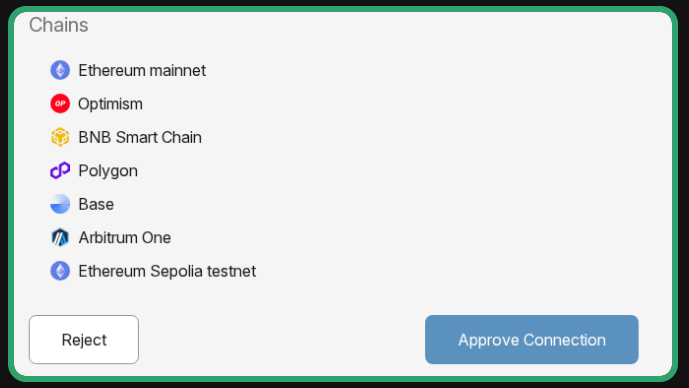
ETH · Arbitrum · BNB | offline | 3/3 identical

WALLETCONNECT · DAPP → BITBOXAPP → DEVICE · APP 3C3644BB2 → 4305EDB6D

pairing

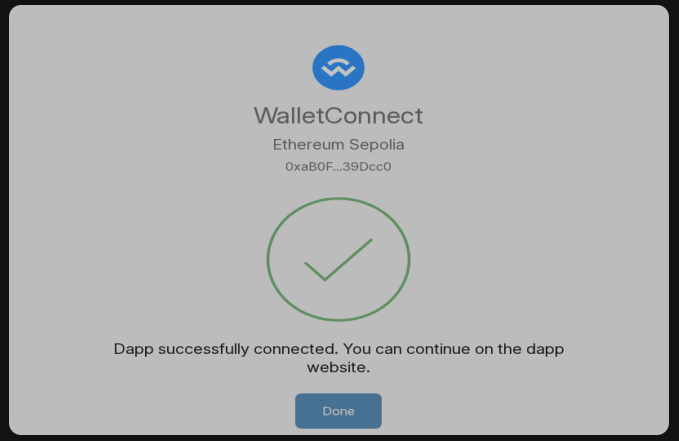


base 5 chains

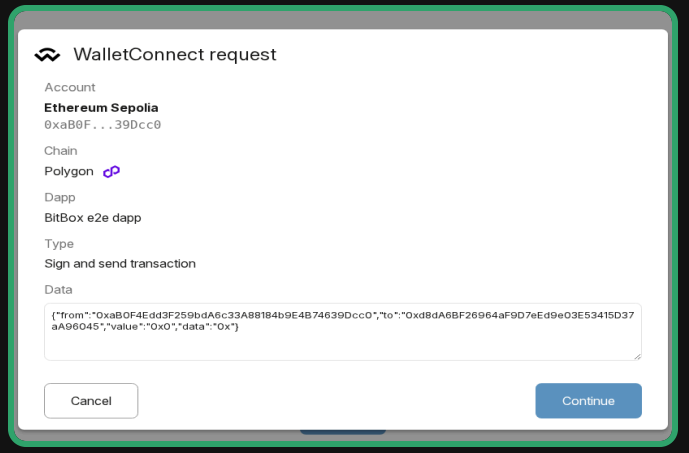


head + BNB, Polygon

send on Polygon



base never reaches the app



head request



head device

EVM CHAINS · 49 DEVICE TESTS · FIRMWARE 061FEC974 ADDS A SCREEN

	address	legacy	1559	erc20	msg	712
ethereum				■		
optimism				■		
bnb				■		
polygon				■		
base				■		
arbitrum				■		
sepolia				■		
123456				■		

changed: erc20, every chain **12 s headless**

2 changed functions → 11 tests selected



optimism



base



arbitrum



chain 123456

SELECTION AND REGRESSION · SAME PR

1 changed function → 28 of 39 tests

eth-erc20-kno…	eth-erc20-unk…	taproot-key-s…	mixed-spend	op-return
multisig-p2wsh	policy-wsh	policy-tr-key…	policy-tr-scr…	taproot-to-no…
send-self-sam…	send-self-dif…	multiple-outp…	multiple-outp…	multiple-outp…
high-fee-roun…	p2wpkh-p2sh	high-input-ad…	locktime-zero	locktime-sequ…
locktime-sequ…	locktime-lite…	locktime-lite…	p2tr-output-m…	op-return-non…
op-return-non…	all-change-hi…	all-change-hi…	all-change-fe…	multisig-not-…
multisig-p2ws…	multisig-large	policy-tr-key…	policy-tr-uns…	policy-tr-uns…
policy-timest…	policy-differ…	policy-wrong-…	policy-change…	

blue: runs `warning_message`, so it ran on the PR · all 28 match the 9.28.0 expected screens

API suites vs PR simulator

repo's own vector copy



firmware's current vectors



pass fail

Found: API repos' vector copies still have the old wording

COVERAGE · NEW FIRMWARE LINES RUN BY THE SUITES (LAST 6 MONTHS)

65%

of 2054 added lines the simulator can reach

Every report lists the changed functions **no test reaches**



BitBox e2e: b59b48983 -> d5fa70250

	base (merge-base)	head
firmware	`b59b48983`	`d5fa70250` (`d5fa70250`)
simulator	bitbox02-multi-v9.28.0-simulator1.0.0-linux-amd64	bitbox02-multi-v9.28.0-simulator1.0.0-linux-amd64
harness	`5fdedb8f7+dirty`	`5fdedb8f7+dirty`

Summary

- **Units**: 28 run (selection: coverage, 1 changed functions), 28 changed (6 distinct), 0 new, 0 with graphical-only differences, 0 failed on head (0 newly)
- **Expectations**: 28 pass, 0 fail, 0 without expectation for this version, 0 units without expectations
- **Unreached changed functions**: none
- **Timings**: baseline 0s, head build 0s, head headless 12s, graphical 40s, lane B 37s

API simulator suites

Committed vectors: the repo's own copy (drift vs. firmware). Firmware vectors: the head firmware's vectors copied in (regression).

suite	head: committed vectors	head: firmware vectors	base: committed	base: firmware
api-go	50 pass / **31 FAIL**	81 pass	81 pass	81 pass
api-rs	52 pass / **1 FAIL**	53 pass	53 pass	53 pass
api-ts	25 pass / **1 FAIL**	26 pass	26 pass	26 pass

▶ api-go (committed vectors): 31 failing, log apis/api-go-committed.log
▶ api-rs (committed vectors): 1 failing, log apis/api-rs-committed.log
▶ api-ts (committed vectors): 1 failing, log apis/api-ts-committed.log

Changed units

btc-vector/taproot-key-spend

WORKFLOW

1

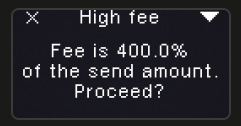
Open PR

`e2e.py pr`

→

2

Report



→

3

Untested code?

Claude drafts a test

→

4

Approve

`e2e.py promote`

→

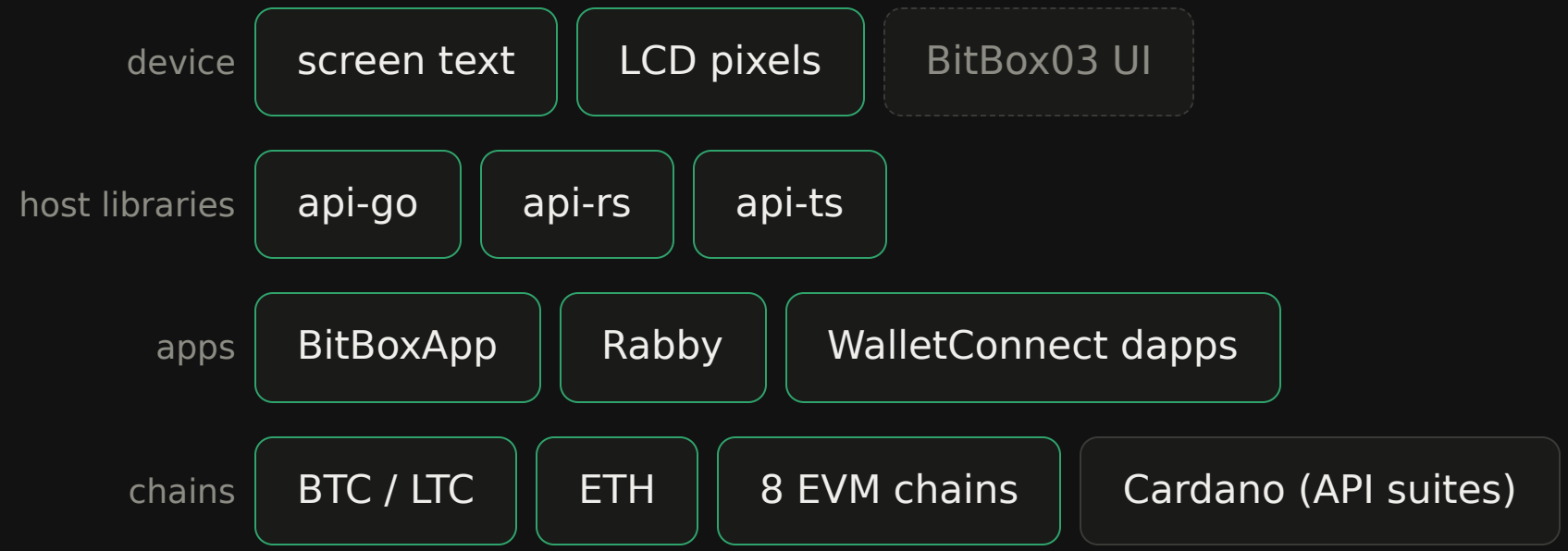
5

Now a check

✓ regression

```
<!-- e2e   optional, in the PR body
units: btc-vector/*fee*
graphical: btc-vector/*fee*
api-rs: my-branch
-->
```

WHAT IT COVERS



solid: running dashed: next